GS EXTERIOR EXPERTS · DECK DIVISION · TAKEOFF + BUILD SET

281 S Jefferson Ave · Decks + Roof Covers · Louisville

281 S Jefferson Ave, Louisville · two decks (12'11" x 11'11" + 14'9" x 3'8" walkway) on CEMCO steel and Diamond Piers · TimberTech Vintage Weathered Teak, picture frame + dividers, no butt joints · two lean-to covers, black standing seam, Hardie soffit / fascia / rakes · blinds kept

SOLD · CHECK OR ACH	COST	GROSS PROFIT	MARGIN	FINANCED	SQUARE FEET
$36,300	**$24,792**	**$11,508**	**31.7%**	**$39,032**	**208**

DESIGN READS

Frame	2 zones, 27'-8" along the house, 11'-11" at the deepest
Deck	51.5 LF of outer edge · 208 SF · 1'-3" above grade
Zone	W Walkway — 14'9" x 3'8" under the small cover: 14'-9" x 3'-8"
Zone	M Main deck — 12'11" x 11'11" under the main cover: 12'-11" x 11'-11"
Joists	CEMCO 600S162-54 C-joist (6", 54 mil / 16 ga, G90) @ 12" OC
Rims	single-ply 600T track rims (G90) and 600T ledger track on a steel angle
Beam	FLUSH boxed 600S162-54 W: 14'-2 1/2", 4 posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 13/16" (worst 473 lb)
Beam	FLUSH boxed 600S162-54 M: 12'-4 1/2", 3 posts at 15'-3" / 21'-2 1/2" / 27'-2" (worst 1,735 lb)
Posts	3 x 6x6 Douglas fir rough deck posts on steel beam saddles + 4 6x6 Douglas fir rough cover posts carrying the deck beam on DP-50/50
Load	50 psf (40 psf live governs) · engineered
Decking	TimberTech Vintage Weathered Teak — 24 rows parallel to the house, picture frame & dividers in Vintage Weathered Teak, 3/16" gaps, CONCEALoc
Rail	none
Stair	main deck side, toward the walkway: 2 risers @ 7.50", 1 treads, 6 stringers, 5'-0" wide

- Rows per panel: panel 1 (0"–14'-9") 7 rows · panel 2 (14'-9"–27'-8") 24 rows · 1 ripped pieces · 1 breaker
- FLUSH boxed 600S162-54 W: 14'-2 1/2" long, 4 posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 13/16"
- FLUSH boxed 600S162-54 M: 12'-4 1/2" long, 3 posts at 15'-3" / 21'-2 1/2" / 27'-2"

- front flush beam IS the front rim — joists hang into it, posts under it, no separate rim plies
- Frame is CEMCO cold-formed steel: the layout is sized as the equivalent-depth lumber, then the engineer of record designs the members (gauge, beam build-up, clip schedule). Steel prices are estimates until the CEMCO dealer quotes.

SITE & CODE FLAGS internal

ENGINEER	ENGINEERING	FRONT FLUSH boxed 600S162-54: boxed 600S162-54: no IRC beam table for steel — laid out as (2)2x8; engineer to confirm IRC Table R507.5
ENGINEER	WIND	basic wind speed 145 mph — uplift on the frame and rail loads; engineered connections (clip angles at every joist end, beam-to-post brackets, H2.5AZ at every cover rafter) ASCE 7
ENGINEER	SNOW	Main cover: a lean-to under the house eave takes sliding snow off the house roof and drift against the wall — ASCE 7 sliding + drift surcharge on the cover ASCE 7-22 7.7 / 7.9
ENGINEER	SNOW	Walkway cover: a lean-to under the house eave takes sliding snow off the house roof and drift against the wall — ASCE 7 sliding + drift surcharge on the cover ASCE 7-22 7.7 / 7.9
ENGINEER	ENGINEERING	CEMCO cold-formed steel frame: no prescriptive deck span table in the IRC for CFS — the engineer of record designs joists, beams, clips and the ledger connection (AISI S240 / S100); member sizes here are the same-depth lumber layout IRC R507 / AISI S240
CODE	SNOW	standing seam roof over the deck — metal sheds its snow load onto the deck and the stair below: snow retention bar the full low eave (S-5! ColorGard or equal, seam clamps, no penetrations), sized for the ground snow ASCE 7 Ch. 7 / manufacturer
CODE	STAIRS	step:W-M stair lands on grade — a landing is required at the bottom (36" min in the direction of travel); pour a pad or set pavers IRC R311.7.6
CODE	ENGINEERING	Louisville amends IRC R507.1: decks supported from the house are engineered unless 30" or less above grade — the decks are under 30", but the covers (IBC App. I patio covers) and the steel frame need the stamped set; the handout asks for every sheet scaled and stamped LMC 15.05.140 / IRC R507.1 amd.
CODE	FIRE	Louisville submittal: decking spec sheet showing ASTM E84 Class A (flame spread 25 or less) — confirm the TimberTech Vintage (Advanced PVC) data sheet lists Class A before submittal (the decking changed from Landmark) Louisville deck handout / Fire Hardening Code
CODE	SITE	under-deck ground: gravel over 6-mil poly (or concrete) where the deck is 6' or less above grade — included under both decks Louisville deck handout
CHECK	FOOTING	Diamond Pier: pins 50" — locate utilities (811) before driving; bearing 473 lb/post vs 3,300 lb allowable (manufacturer chart, confirm)

CHECK	LEDGER	verify the rim board behind the ledger is solid 2x/LVL/engineered rim (not I-joist web or a hollow band) before the ledger fasteners go in IRC R507.9.1.1
CHECK	FOOTING	Main cover: cover post + deck beam share a Diamond Pier — worst ~1,903 lb roof (ground snow 30 + 15 psf dead) + 934 lb deck = ~2,837 lb vs 3,300 lb DP-50/50; uplift at 145 mph — engineer confirms the pier and the post-to-pier hold-down IBC App. I / ASCE 7
CHECK	FOOTING	Walkway cover: cover post + deck beam share a Diamond Pier — worst ~475 lb roof (ground snow 30 + 15 psf dead) + 473 lb deck = ~948 lb vs 3,300 lb DP-50/50; uplift at 145 mph — engineer confirms the pier and the post-to-pier hold-down IBC App. I / ASCE 7
CHECK	PRODUCT	TimberTech Vintage hidden fasteners and picture-frame screws into steel: use the maker's steel-frame screw (self-drilling, for 14–18 ga) — confirm the fastener and that the decking warranty covers steel framing
CHECK	PRODUCT	steel frame prices are estimates (no CEMCO dealer quote yet) — get the take-off quoted by the CEMCO dealer before the contract is signed
CHECK	SETBACK	Dutch Creek PUD (approved 12/15/1981, underlying R-L): the PUD final plan governs setbacks and coverage (LMC 17.28.230) and was not retrievable online. Plain R-L would need 25' rear — the existing house itself sits ~24' from the rear line, and the new main cover's roof edge ~11'-10". Confirm the PUD setbacks with Planning (303-335-4592, Planning@louisvilleco.gov) before submitting; a variance or PUD amendment is outside this scope LMC 17.12.040 / 17.28.230 / Louisville deck handout
CHECK	SETBACK	lot coverage: covered decks count to the support posts. Lot 4,678 SF; house + garage 1,320 SF (assessor). The 2024 aerial puts the existing main cover's roof edge ~11'-4" off the wall; the new cover (posts at 11'-9", 1' overhang) reaches ~12'-9" — about +17 SF of coverage (31.3% to 31.6%; R-L max 30%, the PUD governs). If Planning holds coverage to the existing, set the new posts on the existing post line (field-measure) and let the deck run past the cover Louisville deck handout / LMC 17.12.040
CHECK	SETBACK	work within 3' of a setback may need an Improvement Location Certificate (ILC) — budget a surveyor if Planning asks Louisville deck handout
CHECK	SITE	811 locate before the piers; flood zone X (minimal) Colorado 811 / FEMA
CHECK	FOOTING	Diamond Piers under the cover posts: the city's patio-cover guide asks for footings below the 36" frost line and 8" min. piers — submit the Diamond Pier evaluation report as the frost-protected alternate and get it accepted at plan review Louisville patio-cover guide / IRC R403.1.4 alternate
INFO	SNOW	standing seam on a low-slope shed (~1:12 to 2:12): mechanically seamed or snap-lock rated for the pitch, high-temp self-adhered underlayment under every panel, headwall flashing counterflashed into the house wall
INFO	LEDGER	lateral: 4 tension ties per ledger, steel-joist kit / engineer's detail, into the house floor framing IRC R507.9.2

INFO	STAIRS	step:W-M stair: 2 risers @ 7.50" · 1 treads @ 11.66" · run 11 11/16" · stringers 6 @ 12" OC (composite treads) IRC R311.7.5 / TimberTech
INFO	ROOF	Main cover: 1.03:12 pitch — mechanically seamed (double-lock) panels, rated to 1/2:12 by most makers; high-temp self-adhered underlayment over the whole deck, panels one piece eave to headwall
INFO	ENGINEERING	Main cover: 4x10 beam spans 11'-11" — about 1,184 psi bending vs ~1,200 psi allowable
INFO	ROOF	Walkway cover: 2.05:12 pitch — mechanically seamed (double-lock) panels, rated to 1/2:12 by most makers; high-temp self-adhered underlayment over the whole deck, panels one piece eave to headwall
INFO	ENGINEERING	Walkway cover: (2)2x8 beam spans 5'-4" — about 216 psi bending vs ~1,200 psi allowable
INFO	SETBACK	decks 30" or less above grade are not held to the principal-structure setbacks (Louisville deck handout); both decks are ~15" Louisville deck handout (7/7/2025)
INFO	SNOW	design criteria (Louisville 2024 codes): 145 mph Vult, Exposure C, SDC B (SDS 0.17), frost 36", elev. 5,337', CZ 5B; snow per the ASCE hazard tool (city's Marshall Fire sheet: 30 psf non-reducible; Boulder County layer shows 40 psf) — engineer to confirm the site value 2024 IRC/IBC as adopted, Ord. 1907-2025

THE ORDER Decks & Docks · GSEXT1 · ship to the job

Lumber

NET	ORDER	UNIT	ITEM	WHY
377	**396**	LF	CEMCO 600S162-54 C-joist (6", 54 mil / 16 ga, G90) — joists, PF joists, blocking	+5% (cut to length by the supplier where possible) — 12" OC
79	**84**	LF	CEMCO 600T track (54 mil / 16 ga, G90) — rims (single ply, caps the joist ends) + ledger track	+5% (52 LF rim + 28 LF ledger)
28	**31**	LF	Steel ledger angle 3" x 3" 14 ga G90 — joist seats on the ledger track	+10%
53	**56**	LF	CEMCO 600S162-54 C, G90 — boxed beams (2 C toe-to-toe)	+5% (FLUSH boxed 600S162-54 W 14.2' · FLUSH boxed 600S162-54 M 12.4')
4	**4**	ea	Steel side bracket, 11 ga galvanized — deck beam bolted to the 6x6 cover post (shared pier)	exact (one per shared post, (2) 1/2" thru-bolts + #12 screws into the beam)
1	**1**	ea	6x6x8 Douglas fir rough — deck posts on the deck-only piers	exact (3 posts, about 6" each, 8 per 8')
3	**3**	ea	Simpson ABA66RZ post base on the Diamond Pier bolt — deck post	exact
3	**3**	ea	Galvanized steel beam saddle, 11 ga — boxed steel beam to the 6x6 post top	exact (one per deck post, thru-bolted to the post, #12 screws into the beam)

Footings

NET	ORDER	UNIT	ITEM	WHY
7	**7**	ea	Diamond Pier DP-50/50 (head + 4 x 50" pins, plugs, caps)	exact — confirm stock

Hardware

NET	ORDER	UNIT	ITEM	WHY
60	**66**	ea	Steel clip angle 1-1/2" x 1-1/2" x 7" 16 ga G90 — joist to track / beam	+6 (2 per joist end where it frames into a beam or ledger)
4	**4**	box	#12-14 x 1" hex-washer self-drilling screw, Climaseal, 500 ct	1572 needed (clips, track, boxed beams, post brackets)
1	**1**	box	Structural screw 1/4" x 3-1/2" through the ledger track into the house rim board (or per the engineer)	42 needed — 2 rows staggered 16" OC
8	**8**	ea	Lateral-load tension tie for steel joists (Simpson DTT2Z w/ steel-joist kit or the engineer's detail) — IRC R507.9.2 equivalent	exact (2 near each end of every ledger)
2	**2**	ea	Cold-galvanizing spray (cut ends, field drilling)	lot

Flashing & waterproofing

NET	ORDER	UNIT	ITEM	WHY
1	**1**	roll	Grace Vycor Deck Protector 12" x 75' (wall membrane behind ledger + side rims)	28 LF of house wall with laps — nothing on a privacy wall
3	**3**	ea	Aluminum L-flashing 10' (over ledger + side rims)	house walls only
1	**1**	roll	6-mil poly sheeting 10' x 100' — under the decks	229 SF with laps
1.4	**1.5**	CY	3/4" washed gravel, 2" over the poly — under the decks (delivered)	229 SF x 2"

Decking

NET	ORDER	UNIT	ITEM	WHY
31	**33**	ea	TimberTech Vintage Weathered Teak 1x6x16 Grooved	+2 (31 x 16' · panel 1 7 rows · panel 2 24 rows · every row one piece)
4	**4**	ea	TimberTech Vintage Weathered Teak 1x6x16 Square Edge — borders & dividers	exact (border 14'-9",11'-11" 13'-0" · border 27'-8",0" 12'-0" · breaker DW/M 11'-10" · border 14'-9",3'-8" 8'-4")
1	**2**	ea	TimberTech Vintage Weathered Teak 1x6x20 Square Edge — borders & dividers	+1 (border 0",3'-8" 14'-10" + border 0",0" 3'-9")

Fasteners

NET	ORDER	UNIT	ITEM	WHY
1	**1**	box	TimberTech CONCEALoc hidden fastener 500 ct (screws incl.)	448 needed · 500 supplied
1	**1**	box	Starborn Cap-Tor xd 2-3/4" Epoxy Weathered Teak 350 ct	274 screws needed (first row 30 · border 112 · fascia 84 · stairs 48)

Fascia

NET	ORDER	UNIT	ITEM	WHY
5	**6**	ea	Deck skirt — HardieTrim 4/4 x 12 x 12' smooth ColorPlus Arctic White (wraps the double rim, in place of the composite fascia)	+1 (left fascia (W) 3'-8 1/2" · right fascia (W) 3'-8 1/2" · front fascia (W) 12'-0" · front fascia (W) 2'-10" · left fascia (M) 11'-11 1/2" · right fascia (M) 11'-11 1/2" · front fascia (M) 12'-0" · front fascia (M) 1'-0")
1	**1**	box	Color-matched trim screws for HardieTrim, 100 ct (Arctic White)	40 color-matched screws, pre-drilled, 2 per joist line
1	**1**	kit	ColorPlus touch-up kit, Arctic White	cut ends

Stairs

NET	ORDER	UNIT	ITEM	WHY
1	**2**	ea	TimberTech Vintage Weathered Teak 1x6x12 Square Edge — treads (main deck side, toward the walkway stair)	+1 (1 treads x 2 boards @ 5'-0", 2 per board)
1	**1**	ea	TimberTech Vintage Weathered Teak riser board 7-1/4" x 12' (main deck side, toward the walkway stair)	exact (2 risers @ 5'-0")
2	**2**	ea	Steel stair stringer, galvanized, pre-formed to the riser / tread	exact

Porch cover

NET	ORDER	UNIT	ITEM	WHY
2	**2**	ea	6x6x10 Douglas fir rough post (cover) — Main cover	exact (2 posts on the beam line, field-cut)
1	**1**	ea	4x10x14 Douglas fir — cover beam — Main cover	+0 (4x10 x 13.7' at the posts, one piece)
1	**1**	ea	2x8x16 #1 SYP — cover ledger on the house — Main cover	exact
10	**11**	ea	2x8x14 #1 SYP — cover rafters @ 16" OC — Main cover	+1 cull (10 rafters, 12.5' with the overhang)
3	**4**	ea	2x8x16 #1 SYP — cover fascia / sub-fascia — Main cover	+1 (38 LF)
6	**6**	sheet	7/16" OSB roof sheathing 4x8 — Main cover	171 SF + 10% cuts
171	**180**	SF	Standing seam roof — 24 ga steel 16" 1-1/2" mechanically seamed (double-lock, low-slope rated) panels x 12.5' (Black, PVDF), 10 panels — Main cover	10 panels x 12.5' + 5% (one-piece eave to headwall, no end laps) — mechanical seam priced at the snap-lock rate; confirm with the metal supplier
1	**1**	roll	High-temp self-adhered ice & water underlayment, 2 sq roll (under every metal panel) — Main cover	171 SF — full coverage under metal (high-temp)
2	**3**	ea	Standing seam eave / drip trim 10', color-matched — Main cover	13 LF low eave + 1
3	**4**	ea	Standing seam gable / rake trim 10', color-matched — Main cover	two rakes x 12.5' + 1
2	**3**	ea	Headwall (ledger) flashing with Z-closure 10', color-matched — into the house siding / counterflash — Main cover	13 LF at the house — counterflash into the siding / brick
3	**3**	ea	Foam + metal closure strips at eave and headwall, 10' — Main cover	eave + headwall closures
1	**1**	box	Panel clips + pancake-head screws, box of 250 — Main cover	90 clips @ 18" OC on each seam
1	**1**	box	Color-matched #10 woodgrip trim screws w/ washers, 250 — Main cover	trims, closures, snow bar

NET	ORDER	UNIT	ITEM	WHY
3	**3**	ea	Butyl tape / polyurethane sealant, per roll or tube — Main cover	headwall, eave, rake laps
13	**15**	LF	Snow retention — clamp-on bar across the low eave over the deck (Black) — Main cover	13' — metal sheds 30 psf snow onto the deck and stair below; one bar 12-18" above the eave
5	**5**	panel	Porch ceiling — HardieSoffit 4' x 10' cedarmill ColorPlus Arctic White (non-vented; 1 in 3 vented if the roof is closed) — Main cover	171 SF + 10% cuts
3	**3**	lb	Stainless ring-shank siding nails 2", per lb — Main cover	stainless ring-shank, 6" OC at every rafter
4	**5**	ea	Cover fascia + rakes — HardieTrim 4/4 x 7-1/4 x 12' rustic ColorPlus Arctic White over the 2x8 sub-fascia — Main cover	+1 (38 LF: front 13' + two rakes 12.5')
1	**1**	box	Color-matched trim screws for HardieTrim, 100 ct (Timber Bark) — Main cover	color-matched trim screws
2	**2**	ea	Simpson ABA66RZ post base on the shared Diamond Pier bolt — cover post (pier counted with the deck) — Main cover	exact
2	**2**	ea	Simpson BC46Z post cap, 6x6 post to the 4x beam — sized for a rough 6x6 (R-series), confirm SKU — Main cover	exact
10	**10**	ea	Simpson H2.5AZ rafter tie — Main cover	one per rafter at the beam
20	**20**	ea	FastenMaster LedgerLOK 6" — cover ledger, 2 rows staggered 16" OC — Main cover	13 LF x 2 rows @ 16"
2	**2**	ea	6x6x10 Douglas fir rough post (cover) — Walkway cover	exact (2 posts on the beam line, field-cut)
2	**2**	ea	2x8x16 #1 SYP — cover beam (2 ply) — Walkway cover	+0 ((2)2x8 x 6.3' at the posts)
1	**1**	ea	2x8x16 #1 SYP — cover ledger on the house — Walkway cover	exact
5	**6**	ea	2x8x8 #1 SYP — cover rafters @ 16" OC — Walkway cover	+1 cull (5 rafters, 5.2' with the overhang)
2	**3**	ea	2x8x16 #1 SYP — cover fascia / sub-fascia — Walkway cover	+1 (17 LF)
2	**2**	sheet	7/16" OSB roof sheathing 4x8 — Walkway cover	37 SF + 10% cuts
37	**39**	SF	Standing seam roof — 24 ga steel 16" 1-1/2" mechanically seamed (double-lock, low-slope rated) panels x 5.2' (Black, PVDF), 5 panels — Walkway cover	5 panels x 5.2' + 5% (one-piece eave to headwall, no end laps) — mechanical seam priced at the snap-lock rate; confirm with the metal supplier
1	**1**	roll	High-temp self-adhered ice & water underlayment, 2 sq roll (under every metal panel) — Walkway cover	37 SF — full coverage under metal (high-temp)
1	**2**	ea	Standing seam eave / drip trim 10', color-matched — Walkway cover	6 LF low eave + 1
2	**3**	ea	Standing seam gable / rake trim 10', color-matched — Walkway cover	two rakes x 5.2' + 1
1	**2**	ea	Headwall (ledger) flashing with Z-closure 10', color-matched — into the house siding / counterflash — Walkway cover	6 LF at the house — counterflash into the siding / brick
2	**2**	ea	Foam + metal closure strips at eave and headwall, 10' — Walkway cover	eave + headwall closures
1	**1**	box	Panel clips + pancake-head screws, box of 250 — Walkway cover	20 clips @ 18" OC on each seam

NET	ORDER	UNIT	ITEM	WHY
1	**1**	box	Color-matched #10 woodgrip trim screws w/ washers, 250 — Walkway cover	trims, closures, snow bar
3	**3**	ea	Butyl tape / polyurethane sealant, per roll or tube — Walkway cover	headwall, eave, rake laps
6	**8**	LF	Snow retention — clamp-on bar across the low eave over the deck (Black) — Walkway cover	6' — metal sheds 30 psf snow onto the deck and stair below; one bar 12-18" above the eave
2	**2**	panel	Porch ceiling — HardieSoffit 4' x 10' cedarmill ColorPlus Arctic White (non-vented; 1 in 3 vented if the roof is closed) — Walkway cover	37 SF + 10% cuts
1	**1**	lb	Stainless ring-shank siding nails 2", per lb — Walkway cover	stainless ring-shank, 6" OC at every rafter
2	**3**	ea	Cover fascia + rakes — HardieTrim 4/4 x 7-1/4 x 12' rustic ColorPlus Arctic White over the 2x8 sub-fascia — Walkway cover	+1 (17 LF: front 6' + two rakes 5.2')
1	**1**	box	Color-matched trim screws for HardieTrim, 100 ct (Timber Bark) — Walkway cover	color-matched trim screws
2	**2**	ea	Simpson ABA66RZ post base on the shared Diamond Pier bolt — cover post (pier counted with the deck) — Walkway cover	exact
2	**2**	ea	Simpson AC6RZ post cap, rough 6x6 to the (2)2x8 beam — Walkway cover	exact
5	**5**	ea	Simpson H2.5AZ rafter tie — Walkway cover	one per rafter at the beam
10	**10**	ea	FastenMaster LedgerLOK 6" — cover ledger, 2 rows staggered 16" OC — Walkway cover	6 LF x 2 rows @ 16"

Site

NET	ORDER	UNIT	ITEM	WHY
198	**198**	SF	Demo existing deck + haul-off	labor line — field verify the existing deck

FASTENER & CONNECTOR SCHEDULE

Ledger	600T ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer confirms) · flashing and membrane as for wood
Field	CONCEALoc at every joist in every gap (448)
Face screws	Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then alternate top/bottom every 12", pre-drilled · stair treads 2 per stringer per board
Steel frame	600S162-54 C-joists @ 12" OC · single-ply 600T track rims · boxed 600S162-54 beams · all G90 · clip angles with #12 self-drilling screws (4 per leg) · cut ends cold-galvanized · no joist tape needed

▼ Cut list

MEMBER	STOCK	LENGTH	QTY	NOTE
Field joists (W)	600S162-54	3'-3 1/8"	14	@ 12" OC, crown up

MEMBER	STOCK	LENGTH	QTY	NOTE
Picture-frame joists (W)	600S162-54	3'-3 1/8"	2	centre 5 9/16" from each frame face
Side rim plies (W)	600S162-54	3'-3 1/8"	4	2-ply, laminated on the ground with RSS 2 rows @ 12" staggered
Ledger (W)	600S162-54	14'-9"	1	top at deck height less board thickness; membrane behind, flashing over
Blocking (W)	600S162-54	10 3/8"	15	1 row at 1'-10" from the wall
Field joists (M)	600S162-54	11'-6 1/8"	12	@ 12" OC, crown up
Picture-frame joists (M)	600S162-54	11'-6 1/8"	2	centre 5 9/16" from each frame face
Divider doublers (M)	600S162-54	11'-6 1/8"	1	sister a joist on the divider line so both board edges bear
Side rim plies (M)	600S162-54	11'-6 1/8"	4	2-ply, laminated on the ground with RSS 2 rows @ 12" staggered
Ledger (M)	600S162-54	12'-11"	1	top at deck height less board thickness; membrane behind, flashing over
Blocking (M)	600S162-54	10 3/8"	13	1 row at 5'-11 1/2" from the wall
Field boards panel 1	Vintage Weathered Teak	14'-1"	6	one board wall to border, no joints
Field boards panel 1 (rip)	Vintage Weathered Teak	14'-1"	1	one board wall to border, no joints; ripped to fit the border / the house
Field boards panel 2	Vintage Weathered Teak	12'-2"	17	one board wall to border, no joints
Field boards panel 2	Vintage Weathered Teak	12'-3"	7	one board wall to border, no joints
Stair stringers (main deck side, toward the walkway)	steel stringer	2'-7"	6	2 risers @ 7.500" · 1 treads @ 11.66" · stringers 12" OC
FLUSH boxed 600S162-54 W beam	600S162-54	14'-2 1/2"	2	y 3'-6 3/8"; posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 13/16"
6x6 deck posts / cover posts under FLUSH boxed 600S162-54 W	6x6 DF rough	6"	4	field-measure each
FLUSH boxed 600S162-54 M beam	600S162-54	12'-4 1/2"	2	y 11'-9 3/8"; posts at 15'-3" / 21'-2 1/2" / 27'-2"
6x6 deck posts / cover posts under FLUSH boxed 600S162-54 M	6x6 DF rough	6"	3	field-measure each
Border 27'-8",0"	Vintage Weathered Teak square	12'-0"	1	mitre at the corners; dividers on the doubled joist
Border 14'-9",11'-11"	Vintage Weathered Teak square	13'-0"	1	mitre at the corners; dividers on the doubled joist
Border 14'-9",3'-8"	Vintage Weathered Teak square	8'-4"	1	mitre at the corners; dividers on the doubled joist
Border 0",3'-8"	Vintage Weathered Teak square	14'-10"	1	mitre at the corners; dividers on the doubled joist

MEMBER	STOCK	LENGTH	QTY	NOTE
Border 0",0"	Vintage Weathered Teak square	3'-9"	1	mitre at the corners; dividers on the doubled joist
Breaker dw/m	Vintage Weathered Teak square	11'-10"	1	mitre at the corners; dividers on the doubled joist
Stair treads (main deck side, toward the walkway)	Vintage Weathered Teak square	5'-0"	2	2 boards per tread, nosing 3/4" over the riser
Stair risers (main deck side, toward the walkway)	riser board	5'-0"	2	rip to 7.50" less tread thickness
Left fascia (w)	fascia	3'-8 1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia (w)	fascia	3'-8 1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Front fascia (w)	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Front fascia (w)	fascia	2'-10"	1	flush under the board nose; full board at the front corners, short piece at the house
Left fascia (m)	fascia	11'-11 1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Right fascia (m)	fascia	11'-11 1/2"	1	flush under the board nose; full board at the front corners, short piece at the house
Front fascia (m)	fascia	12'-0"	1	flush under the board nose; full board at the front corners, short piece at the house
Front fascia (m)	fascia	1'-0"	1	flush under the board nose; full board at the front corners, short piece at the house

THE STACK — COST + MARGIN sold at $36,300 · internal

Materials (raw)	$13,553.30
Materials × 1.15	$15,586.29
Sales tax 4.00% on factored materials	$623.45
Labor (rate card)	$7,750.00
Work — carries the margin	$23,959.75
General conditions at cost	$832.00
Cost	$24,791.75
SOLD — check or ACH	**$36,300** · $174.52/SF
Sell at the 40% floor	$40,765
Financed	**$39,032** · $453/mo at 6.99% × 10 yr
Gross profit	$11,508
Margin on the sold price (gross profit ÷ sell, GC at cost)	**31.7%**

Margin on the work (the GM the engine prices with)	32.4%
Markup on cost	46.4%
Commission 10% of sell	$3,630
Gross profit after commission	**$7,878** · 21.7% of sell
Engineering — outside the price, at cost	$1,800–$3,000

Labor

LINE	QTY	UNIT	RATE	EXT
Demolition — existing deck	198	SF	$1.31	$259.34
Diamond Pier install (replaces caisson)	7	pier	$32.75	$229.22
Frame	208	SF	$6.55	$1,362.21
Decking & fascia	208	SF	$5.24	$1,089.77
Stairs (step:W-M)	2	riser	$98.24	$196.47
Porch cover — both roof covers: posts, ledger, beam, rafters, sheathing, roofing, Hardie, set on the piers	1	lot	$4,000.00	$4,000.00
Porch cover (Main cover) — tear off the existing roof cover and haul	171	SF	$1.96	$335.96
Porch cover (Walkway cover) — tear off the existing roof cover and haul	37	SF	$1.96	$72.69
Under-deck prep — 6-mil poly + 2" gravel	208	SF	$0.98	$204.34

General conditions — at cost, no margin

ITEM	AMOUNT	WHY
General conditions	$832	$4/SF x 208 SF

What the price includes — sell values

ITEM	VALUE
Demolition & haul-off	$384
Foundation & structure	$9,405
Decking & fascia	$8,971
Stairs	$971
2 roof covers — lean-to roofs on 6x6 Douglas fir posts, standing seam steel roof with snow retention, Hardie soffit, fascia and rakes, existing covers torn off	$15,737
Site & project services	$832
Total, check or ACH	**$36,300**
Total, financed	**$39,032**

THE QUOTE homeowner-facing · facts only

QUOTE · PRE-ENGINEERING

281 S JEFFERSON AVE — TWO DECKS + TWO LEAN-TO COVERS (CEMCO STEEL, TIMBERTECH VINTAGE WEATHERED TEAK)

CEMCO cold-formed steel frame, all G90 — 600S162-54 C-joists at 12" OC, 600T track rims, boxed 600S162-54 beams — on 7 Diamond Pier footings. TimberTech Vintage Weathered Teak decking, 2 lean-to roof covers with a Black standing seam roof.

208.0 SF · 2 zones · **1'-3"** above grade, engineered · **30 psf** snow design

Prepared for the owner of 281 S Jefferson Ave, Louisville

PRICE

Total investment $39,032 — financed, no money down

- 12 months no payments, no down payment
- 18 months no payments, no down payment
- 6.99% for 10 years — about $453/month

Check or ACH: $36,300 (7% savings)

WHAT THE PRICE INCLUDES

ITEM	VALUE
Demolition & haul-off	$384
Foundation & structure	$9,405
Decking & fascia	$8,971
Stairs	$971
2 roof covers — lean-to roofs on 6x6 Douglas fir posts, standing seam steel roof with snow retention, Hardie soffit, fascia and rakes, existing covers torn off	$15,737
Site & project services	$832
Total, check or ACH	**$36,300**
Total, financed	**$39,032**

Engineering — $1,800 to $3,000. Stamped drawings by a Colorado engineer (30 psf snow). GSX provides the engineer and bills the engineering at cost, with no markup. You own the stamped set. It is separate from the deck price above.

INCLUDED — WHAT IT IS

ELEMENT	SELECTION
Decking	TimberTech PVC Vintage, Weathered Teak; picture frame and dividers in Vintage Weathered Teak. Grooved boards, 3/16" gaps, a HardieTrim skirt around the frame (Arctic White). Ignition Resistant, Class A, WUI; Limited Lifetime product / 50-yr fade & stain.
Frame	CEMCO 600S162-54 C-joist (6", 54 mil / 16 ga, G90) @ 12" OC; single-ply 600T track rims (G90) and 600T ledger track on a steel angle; FLUSH boxed 600S162-54 W: 14'-2 1/2", 4 posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 13/16" (worst 473 lb); FLUSH boxed 600S162-54 M: 12'-4 1/2", 3 posts at 15'-3" / 21'-2 1/2" / 27'-2" (worst 1,735 lb).

ELEMENT	SELECTION
Foundation	7 Diamond Pier foundations (DP-50/50) — no excavation.
Design load	30 psf snow, stamped by a Colorado engineer.
Hardware	Galvanized steel clip angles and self-drilling screws throughout the frame; TimberTech CONCEALoc hidden fasteners in the field and color-matched plugs on the picture frame, every screw the steel-frame type.
Roof covers	2 lean-to covers replaced: 6x6 posts on the Diamond Pier foundations, a new ledger on the house, 2x8 rafters at 16", Black standing seam steel roof, 24 ga, mechanically seamed for the low pitch, over high-temp underlayment, with a snow bar across the low eave; James Hardie soffit ceiling, HardieTrim fascia and rakes in Arctic White. Main cover: 96" clear at the house to 84" at the beam; Walkway cover: 89" clear at the house to 82" at the beam.
Site	The existing decks and roof covers removed and hauled off; 6-mil poly and 2" of gravel under the decks.
Not included	Electrical — no lighting, fans or outlets; Roll-down blinds — existing, they stay; Grading beyond the under-deck prep; Variance or PUD amendment, if Planning requires one.

PLAN

2 zones · 208.0 SF · 27'-8" along the house · 1'-3" above grade

- **W · Walkway — 14'9" x 3'8" under the small cover** — 14'-9" x 3'-8".
- **M · Main deck — 12'11" x 11'11" under the main cover** — 12'-11" x 11'-11".

Zones are named left to right standing in the yard facing the house.

- Stair main deck side, toward the walkway: 2 risers @ 7.50", 1 treads, 6 stringers, 5'-0" wide

SCHEDULE

1. Engineering & ordering (2–3 weeks). Stamped drawings, materials ordered, samples approved.
2. Day 1 — protection, utilities located (811), old deck out.
3. Day 2 — layout, 7 Diamond Piers driven, ledger set.
4. Days 3–4 — posts and beams set.
5. Days 5–6 — frame, flashing at every wall.
6. Days 7–8 — decking, borders first.
7. Day 9 — stairs.
8. Day 10 — final clean; end of construction.

NEXT STEP

Engineering first. This is a quote: the numbers depend on the stamped engineering, so the first step is engineering. GSX provides the Colorado engineer and supplies the design package; you pay GSX the engineer's cost at no markup and you own the stamped set. When the stamped set is in hand, GSX issues a fixed-price proposal on these numbers with the payment terms.

Quote, not a contract: a fixed-price proposal follows the stamped engineering. Quote numbers hold 30 days. Renders and plan are design intent; the stamped engineering set governs structure, dimensions and locations.

RENDERS 3D model · design intent · frame shown as specified



From the yard



From the front corner



Standing on the deck



Under the deck — posts, beams, footings

SUN AND SHADE the real sun at this address · the covers and the blinds through the day



Jul 15 · 9 AM — sun E, 35° up



Jul 15 · 12 PM — sun SE, 67° up



Jul 15 · 3 PM — sun WSW, 60° up



Jul 15 · 6 PM — sun W, 26° up



Dec 21 · 3 PM — sun SW, 14° up

ISOMETRICS



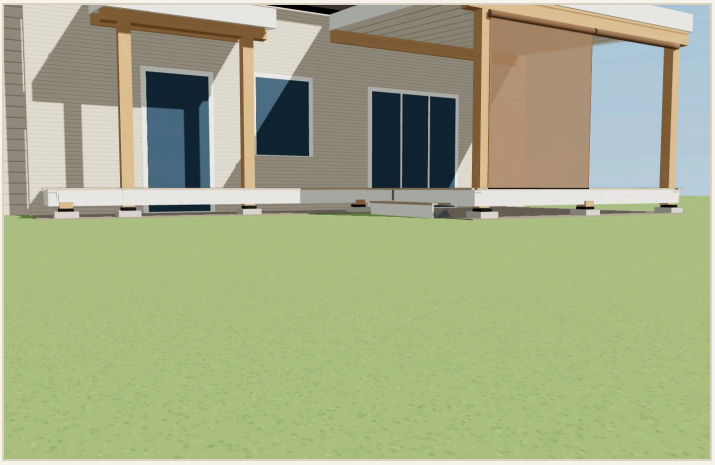
Isometric from the stair corner



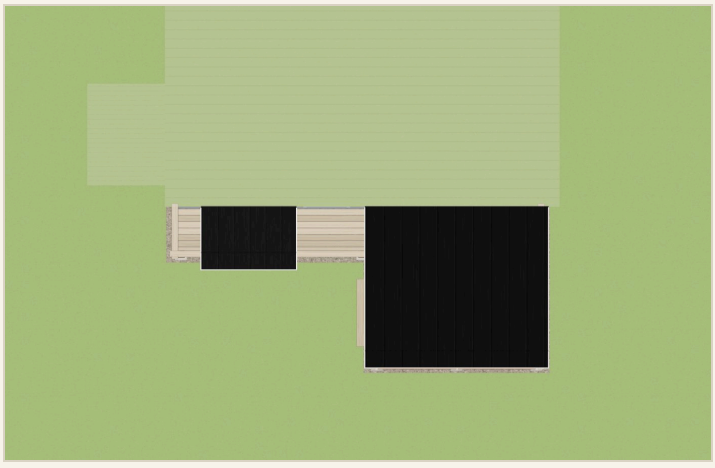
Isometric from the open corner



Low isometric

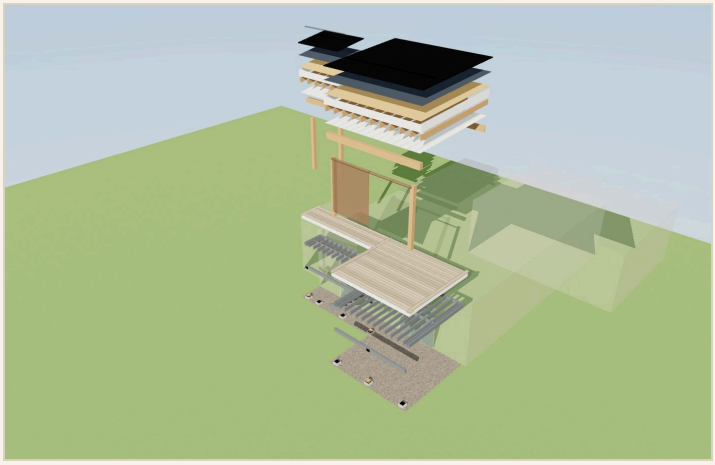


Isometric from under the frame

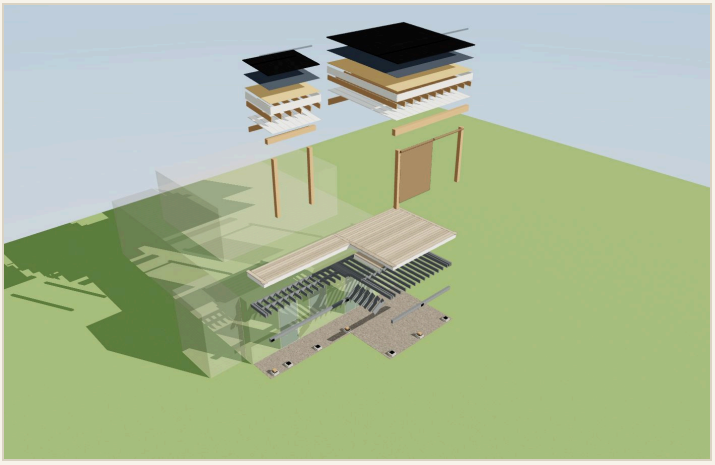


Plan from above

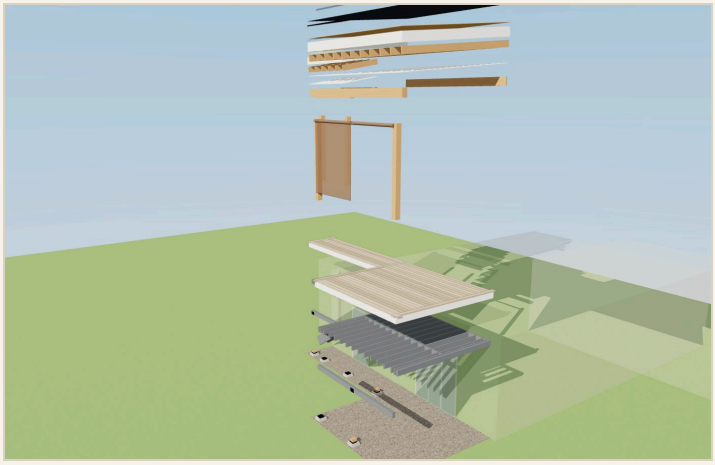
EXPLODED



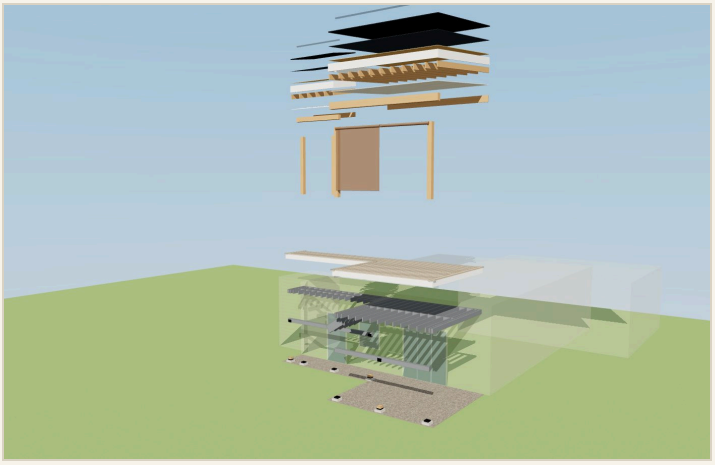
Exploded — every layer



Exploded — from the open corner



Exploded — from the front corner



Exploded — low angle

G-001 General notes · design criteria

G-001 GENERAL NOTES · DESIGN CRITERIA

Loads, codes, structure summary, sheet index, site — the permit-set cover

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

GSX · Jade Pender 303-550-9558 · October 19, 2026 · Rev 0 FOR REVIEW

PROJECT

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel, TimberTech Vintage Weathered Teak)
281 S Jefferson Ave, Louisville CO 80027
208 SF · 1'-3" above grade · 2 zones

DESIGN CRITERIA

Deck live load	40 psf uniform (IRC Table R301.5); guards 200 lb concentrated / 50 plf
Ground snow	30 psf (ground; deck design uses the larger of snow and live) — 40 psf live governs; ground snow per the ASCE hazard tool (Louisville criteria) — 30 psf used here (city sheet, non-reducible); Boulder County layer shows 40 psf: engineer confirms
Dead load	10 psf deck · 15 psf porch cover (standing seam steel on sheathing, Hardie soffit; 12 psf actual, 15 used) carried on the deck posts
Roof snow (cover)	30 psf flat-roof design snow, sliding snow off the house roof + drift at the house wall per ASCE 7 Ch. 7
Design total	50 psf on the deck · cover posts add roof snow + dead to the deck frame and footings (engineered)
Wind	145 mph Vult (3-second gust), Exposure C (Louisville 2024 design criteria), Risk Cat. II
Frost depth	3'-0" (footings bear below)
Soil bearing	1,500 psf presumptive (IRC R401.4.1) — no soils report
Seismic	B
Wildfire (WUI)	not in a WUI zone
Codes	2024 IRC / IBC as adopted and amended by the City of Louisville (Ord. 1907, Series 2025); IRC R507 decks with LMC 15.05.140; IBC Appendix I patio covers; ASCE 7-22 loads; Louisville Fire Hardening provisions; cold-formed steel frame per AISI S240, engineered
Engineering	Stamped structural set by a Colorado PE required — the stamped set governs

STRUCTURE

Footings	DP-50/50
Posts	3 x 6x6 Douglas fir rough deck posts
Beams	FLUSH boxed 600S162-54 W: 14'-2 1/2", 4 posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 13/16" (worst 473 lb); FLUSH boxed 600S162-54 M: 12'-4 1/2", 3 posts at 15'-3" / 21'-2 1/2" / 27'-2" (worst
Joists	CEMCO 600S162-54 C-joist (6", 54 mil / 16 ga, G90) @ 12" OC
Rims	single-ply 600T track rims (G90) and 600T ledger track on a steel angle
Ledger	600T ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer confirms) · flashing and membrane as for wood
Decking	TimberTech Vintage Weathered Teak — 24 rows parallel to the house, picture frame & dividers in Vintage Weathered Teak, 3/16" gaps, CONCEALoc
Rail	none

SHEET INDEX

G-001	General notes · design criteria
C-100	Site plan
A-001	Isometric · renders
A-002	Isometric — assembled, labeled
A-003	Isometric — exploded, labeled
S-100	Foundation plan
S-101	Framing plan
S-102	Structure isometric, labeled
A-101	Decking plan
A-202	Elevations
A-301	Section + front elevation
A-401	Roof covers — plan + section
A-402	Roof covers — exploded isometric
D-501	Details
D-502	Connection details — steel frame
D-503	Connection details — roof covers
T-400	Build steps (build set)
T-700	Order + schedules (takeoff)

SITE

Rear / front lot line: deck edge 12.0' from the line, setback per the PUD...
Side lot line: deck edge 9.8' from the line, setback per the PUD —...

FLAGS (INTERNAL — remove before the permit submittal)

ENGINEER	FRONT FLUSH boxed 600S162-54: boxed 600S162-54: no IRC beam table...
ENGINEER	basic wind speed 145 mph — uplift on the frame and rail loads;...
ENGINEER	Main cover: a lean-to under the house eave takes sliding snow off...
ENGINEER	Walkway cover: a lean-to under the house eave takes sliding snow...
ENGINEER	CEMCO cold-formed steel frame: no prescriptive deck span table in...
CODE	standing seam roof over the deck — metal sheds its snow load onto...
CODE	step:W-M stair lands on grade — a landing is required at the...
CODE	Louisville amends IRC R507.1: decks supported from the house are...
CODE	Louisville submittal: decking spec sheet showing ASTM E84 Class A...
CODE	under-deck ground: gravel over 6-mil poly (or concrete) where the...

No scale

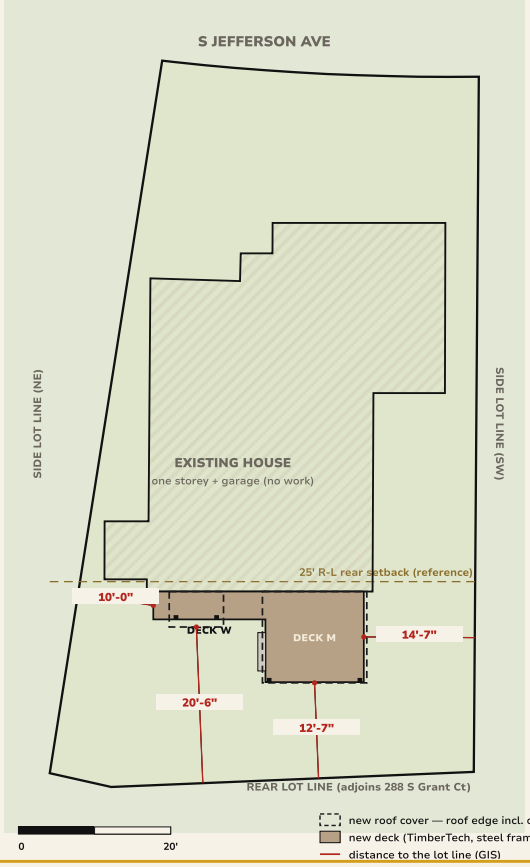
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

C-100 SITE PLAN

Parcel, existing house, new decks and roof covers, distances to the lot lines, setbacks

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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PARCEL

281 S Jefferson Ave, Louisville CO 80027
Parcel 157517119033 · Account R0092293
Lot 33, Block 10, Dutch Creek (plat rec. 00492001, 1982)
Section 17, T1S, R69W
Zoning: R-L within the Dutch Creek PUD (approved 12/15/1981)
Lot 4,678 SF · flood zone X
House 1982 one-storey ranch: 920 SF main floor + 400 SF garage (assessor)

LOT COVERAGE

Lot area	4,678 SF
House + garage (assessor footprint)	1,320 SF
Existing covers to the posts (2024 aerial, est.)	~142 SF (main ~12' x 10'-4", walkway ~5'-4" x 3'-5")
Existing coverage	~1,462 SF · 31.3%
Main cover, new (to the posts) 12'-11" x 11'-8"	151 SF
Walkway cover, new (to the posts) 5'-4" x 3'-5"	18 SF
Proposed coverage	~1,489 SF · 31.8% (+~27 SF)
Uncovered decks	not counted (under 30" above grade)

SETBACKS

Decks are 30" or less above grade — not held to the principal setbacks (city deck handout)
Roof covers: part of the principal structure — they replace the two existing covers in place; the 2024 aerial shows the existing main roof ~1'-5" shallower: field-verify the existing post line
R-L reference setbacks (LMC 17.12.040) — the Dutch Creek PUD plan governs; Planning to confirm
Measured (red): to the nearest lot line from the new work — GIS, not survey

SITE WORK

· Remove the two decks, the two roof covers and the box step; haul off
· Diamond Pier foundations (no excavation spoils), 6-mil poly + 2" gravel under the decks
· Stair lands on grade (no pad)
· Roof drainage: covers shed to grade at the low eave, away from the house
· Call 811 before the piers; protect the lawn and the owner's blinds

Boulder County GIS parcel + 2024 Pictometry roof outline, Louisville GIS zoning / PUD layers. GIS lines are not a survey — an ILC governs if Planning requires one.

Plan · approx. 1" = 20' at full size · GIS lines, not a survey

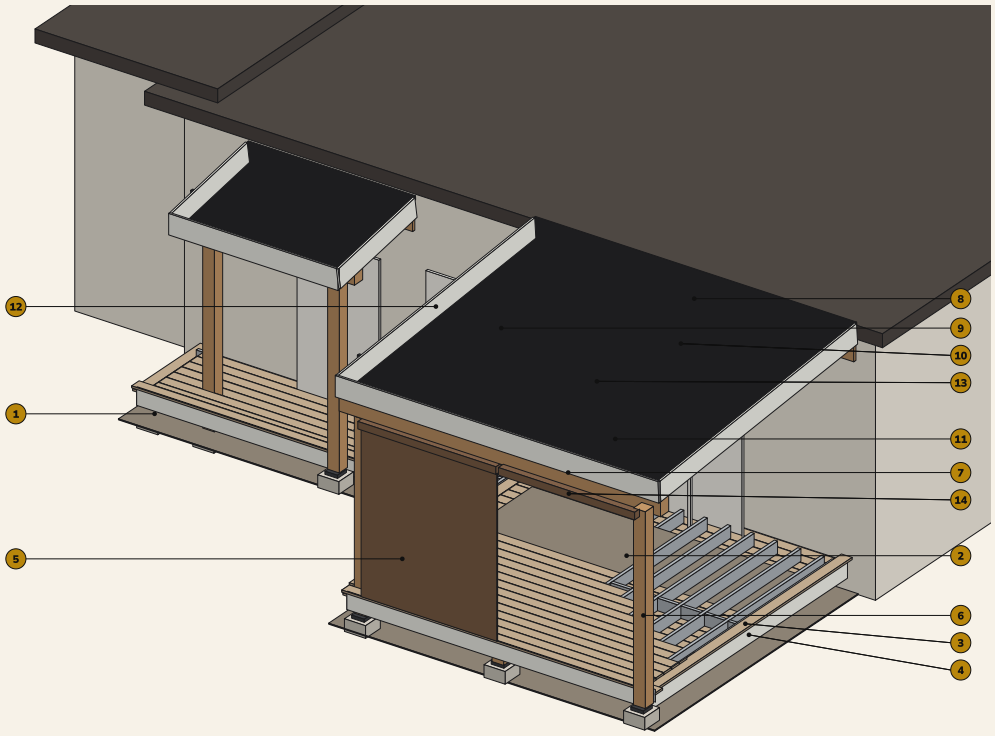
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

A-002 ISOMETRIC — ASSEMBLED, LABELED

Everything you see called out: piers, decking, picture frame, skirt, stair, roof covers, finishes

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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KEY

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. TimberTech Vintage Weathered Teak field decking, hidden fasteners
3. Picture frame + breaker boards — TimberTech Vintage Weathered Teak
4. HardieTrim skirt Arctic White
5. Stair — steel stringers, Vintage treads, closed risers, lands on grade
6. 6x6 Douglas fir rough cover post on ABA66RZ (shared Diamond Pier)
7. 4x10 cover beam, Douglas fir, BC46Z cap
8. 2x8 cover ledger, wood, LedgerLOK 6" 2 rows, headwall flashed
9. 2x8 rafters @ 16" OC, wood, H2.5AZ at the beam
10. Black standing seam, 24 ga, mechanically seamed
11. Snow retention bar, clamped to the seams
12. HardieTrim fascia + rakes Arctic White
13. HardieSoffit cedarmill ceiling Arctic White
14. Existing roll-down blinds (stay)

House shown for context only (no work). The CEMCO steel deck frame under the decking is called out on S-102 and A-003; cover framing is wood: 2x8 rafters, main cover 4x10 beam, walkway cover (2)2x8 beam, 6x6 Douglas fir posts.

Axonometric · not to scale · the model the plans are drawn from

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

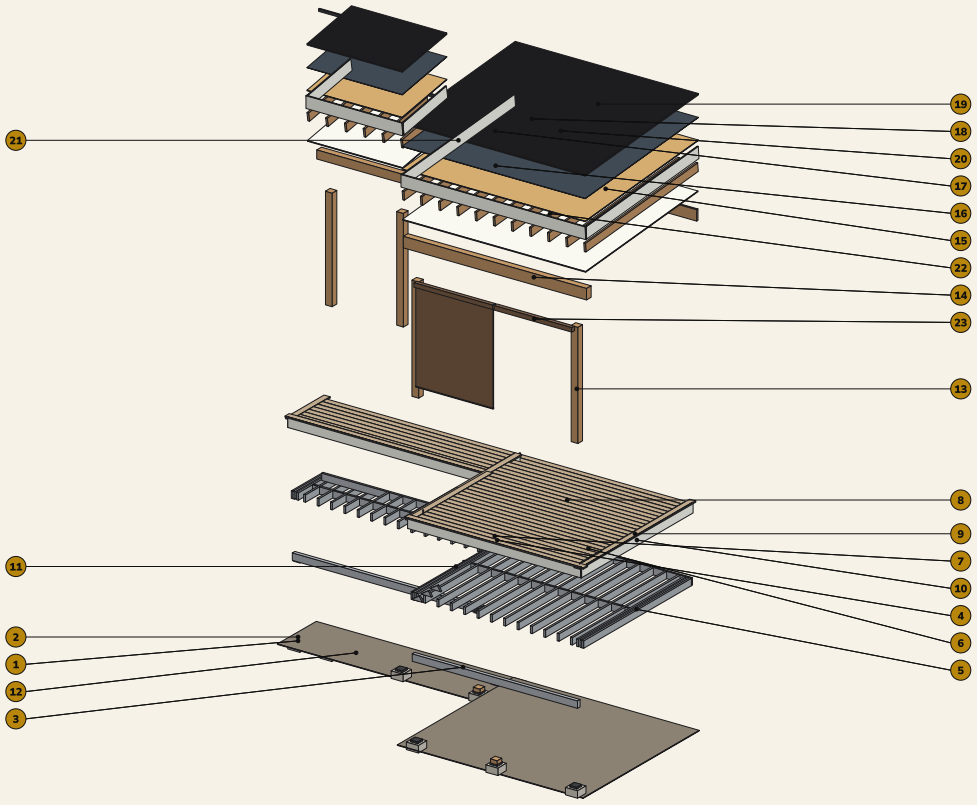
A-003 Isometric — exploded, labeled

A-003 ISOMETRIC — EXPLODED, LABELED

Each build layer lifted in order: piers · posts · beams · frame · decking · stair · roof covers layer by layer

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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KEY

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 6x6 Douglas fir rough deck post on ABA66RZ, steel saddle to the boxed beam
3. Boxed CEMCO 600S162-54 beam (2 C toe-to-toe, G90), flush
4. CEMCO 600S162-54 C-joist, G90, @ 12" OC — clip angles each end
5. CEMCO 600T track rim, G90 (caps the joist ends)
6. Steel ledger track + 3x3 ledger angle on the house rim, flashed
7. Steel blocking at the breaker / picture-frame joists
8. TimberTech Vintage Weathered Teak field decking, hidden fasteners
9. Picture frame + breaker boards — TimberTech Vintage Weathered Teak
10. HardieTrim skirt Arctic White
11. Stair — steel stringers, Vintage treads, closed risers, lands on grade
12. Gravel over 6-mil poly under the decks
13. 6x6 Douglas fir rough cover post on ABA66RZ (shared Diamond Pier)
14. 4x10 cover beam, Douglas fir, BC46Z cap
15. 2x8 cover ledger, wood, LedgerLOK 6" 2 rows, headwall flashed
16. 2x8 rafters @ 16" OC, wood, H2.5AZ at the beam
17. 7/16" OSB roof sheathing
18. High-temp self-adhered underlayment
19. Black standing seam, 24 ga, mechanically seamed
20. Snow retention bar, clamped to the seams
21. HardieTrim fascia + rakes Arctic White
22. HardieSoffit cedarmill ceiling Arctic White
23. Existing roll-down blinds (stay)

Layers lifted in build order. Roof covers pulled apart: posts, beam + ledger, rafters, soffit, sheathing, underlayment, standing seam, snow bar, Hardie trim.

Axonometric · not to scale · the model the plans are drawn from

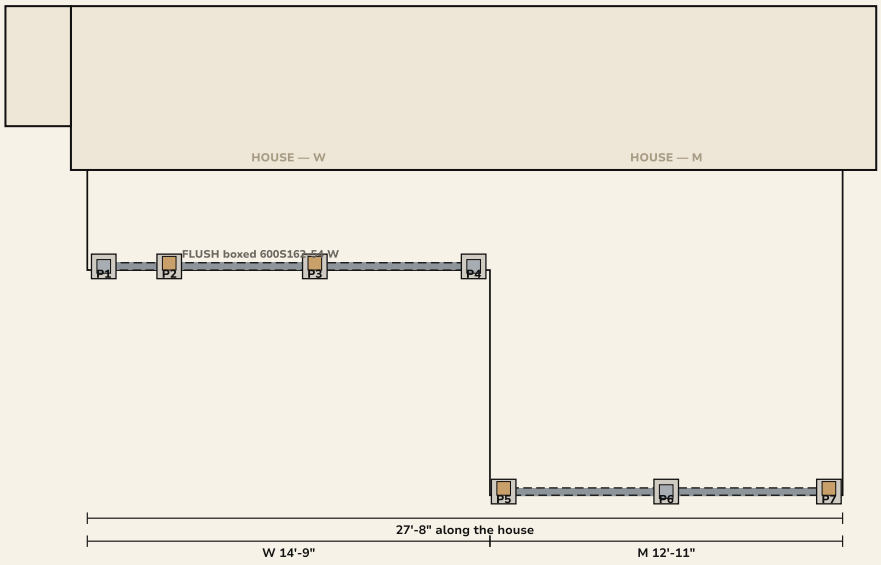
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

S-100 FOUNDATION PLAN

Footings, posts and beam lines with the footing schedule; beams dashed (below the joists)

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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FOOTING SCHEDULE

Diamond Pier DP-50/50 (head + 4 x 50" pins), 3,300 lb allowable

ID	X	Y	ZONE	LOAD
P1	7 3/16"	3'-6 3/8"	W	130 lb
P2	3'-0"	3'-6 3/8"	W +cover	328 lb +roof
P3	8'-4"	3'-6 3/8"	W +cover	473 lb +roof
P4	14'-1 13/16"	3'-6 3/8"	W	275 lb
P5	15'-3"	11'-9 3/8"	M +cover	934 lb +roof
P6	21'-2 1/2"	11'-9 3/8"	M	1,735 lb
P7	27'-2"	11'-9 3/8"	M +cover	934 lb +roof

x from the left frame face facing the house; y from the reference wall.
Frost 3'-0" · soil 1,500 psf · worst post 1,735 lb

Deck-only piers: 6x6 Douglas fir rough post on ABA66RZ, steel saddle to the bea
+cover: the 6x6 Douglas fir rough roof-cover post stands on this pier (ABA66RZ) and carries the deck beam on a steel side bracket (D-502 / 4); roof load adds — the engineer sizes the pier.
Beams: boxed CEMCO 600S162-54, flush (D-502 / 2).

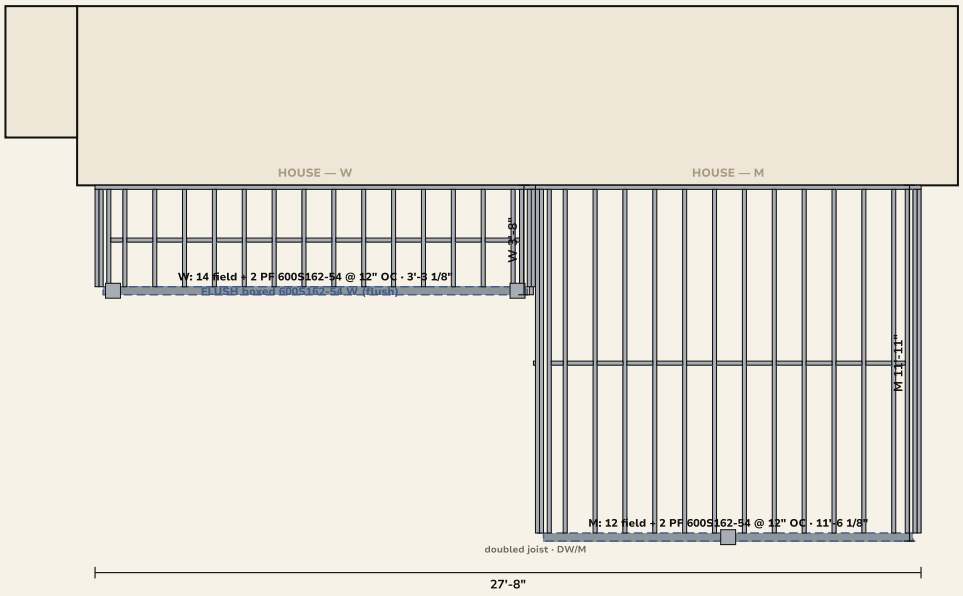
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

S-101 FRAMING PLAN

Ledger, joists, rims, blocking, beams; every member in the model — the cut list is on T-600

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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FRAMING NOTES

CEMCO 600S162-54 C-joist (6", 54 mil / 16 ga, G90) @ 12" OC

single-ply 600T track rims (G90) and 600T ledger track on a steel angle

600T ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer

blocking one row over the beam, staggered

FLUSH boxed 600S162-54 W: 14'-2 1/2", 4 posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 13/16" (worst 473 lb); FLUSH boxed

Connectors:

Steel clip angle 1-1/2" x 1-1/2" x 7" 16 ga G90 — joist to track / beam: 60

Lateral-load tension tie for steel joists (Simpson DTT2Z w/ steel-joist kit or the engineer's detail) — IRC R507.9.2

Cold-galvanizing spray (cut ends, field drilling): 2

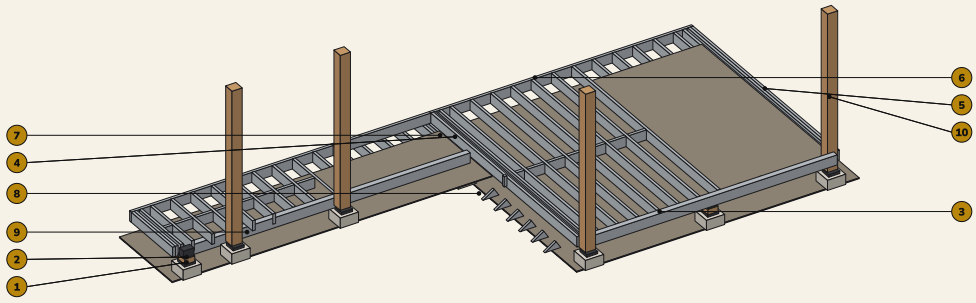
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

S-102 STRUCTURE ISOMETRIC — STEEL FRAME, LABELED

Piers, stubs, cover posts at the shared piers, boxed beams, C-joists, track rims, ledger track, blocking

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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KEY

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 6x6 Douglas fir rough deck post on ABA66RZ, steel saddle to the boxed beam
3. Boxed CEMCO 600S162-54 beam (2 C toe-to-toe, G90), flush
4. CEMCO 600S162-54 C-joist, G90, @ 12" OC — clip angles each end
5. CEMCO 600T track rim, G90 (caps the joist ends)
6. Steel ledger track + 3x3 ledger angle on the house rim, flashed
7. Steel blocking at the breaker / picture-frame joists
8. Stair — steel stringers, Vintage treads, closed risers, lands on grade
9. Gravel over 6-mil poly under the decks
10. 6x6 Douglas fir rough cover post on ABA66RZ (shared Diamond Pier)

Decking, covers and house removed to show the frame. Every joist end clips to the track, ledger track or boxed beam; the engineer's clip schedule governs.

Axonometric · not to scale · the model the plans are drawn from

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

A-101 DECKING PLAN

Every board, border and divider as it lays; rail posts shown for the pockets

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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DECKING

Field: TimberTech Vintage Weathered Teak, 5.5" x 1", 3/16" gaps

Border & dividers: Vintage Weathered Teak

Boards run parallel to the house; edge overhang 1" over the fascia

CONCEALoc at every joist in every gap (448)

Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then

Rows per panel: panel 1 (0"–14'-9") 7 rows · panel 2 (14'-9"–27'-8") 24 rows · 1 ripped pieces · 1 breaker — every board one piece wall to border, no butt

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

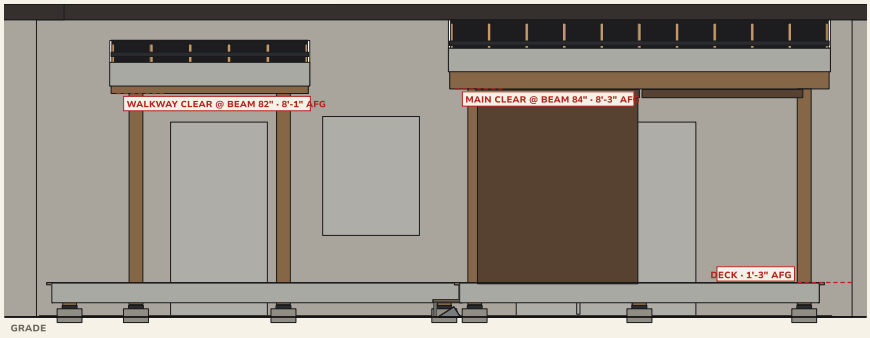
A-202 ELEVATIONS

Front from the yard and both ends, drawn from the model — deck and roof-cover heights

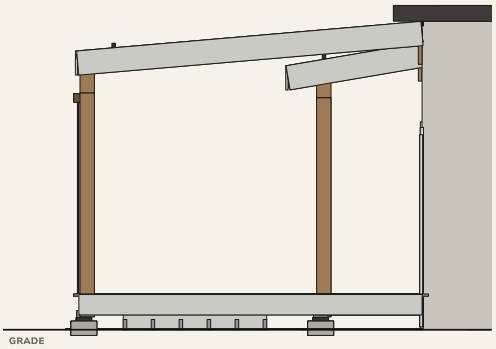
281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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FRONT ELEVATION — from the yard



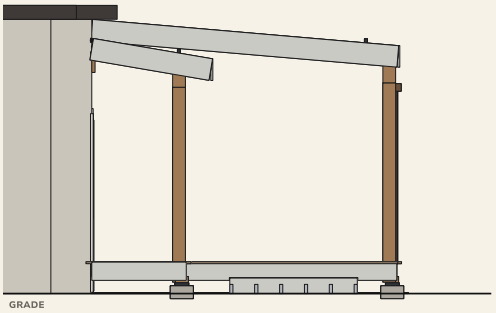
RIGHT END ELEVATION



NOTES

· Deck surface 1'-3" above grade (walkway low point 9")
· Heights AFG = above finished grade at the beam line; field-verify grade before setting the piers
· Clear heights are to the underside of the beam / ledger above the deck surface
· Roof covers: black standing seam on 2x8 rafters, Hardie fascia / rakes / soffit; slope to the yard
· Deck frame: CEMCO cold-formed steel; HardieTrim skirt covers the track rims
· No guard required: every walking surface is under 30" above grade (IRC R312.1)

LEFT END ELEVATION



Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

A-301 Section + front elevation

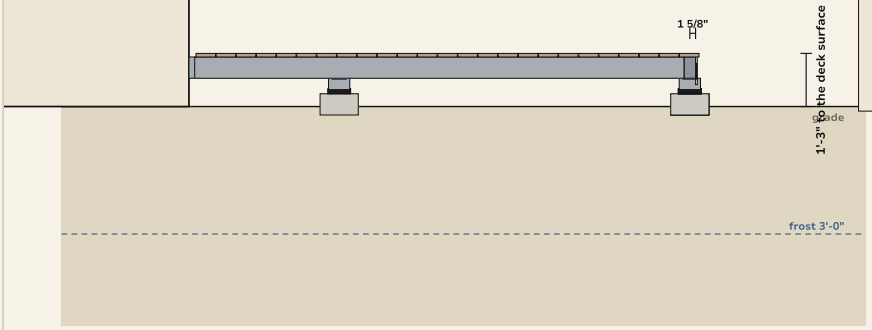
A-301 SECTION + FRONT ELEVATION

Cut through the deepest zone; heights to grade, frost line, cantilever, guard

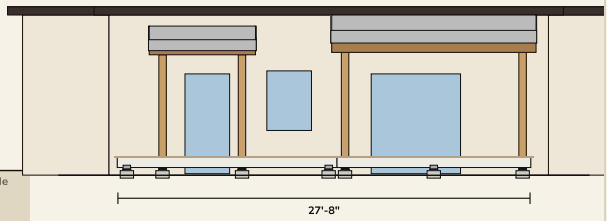
281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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SECTION through zone M at 6'-5 1/2" — looking toward the left end



FRONT ELEVATION — from the yard



SECTION NOTES

Deck surface 1'-3" above grade

FLUSH boxed 600S162-54 W: 14'-2 1/2", 4 posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 13/16" (worst 473 lb); FLUSH boxed 600S162-54 M: 12'-4 1/2", 3 posts at 15'-3" / 21'-2 1/2" /

Posts 3 x 6x6 Douglas fir rough deck posts on steel beam saddles + 4 6x6 Douglas fir rough cover posts carrying the deck beam on DP-50/50

600T ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer confirms) · flashing and membrane as for wood

Section left · elevation right · not to a common scale

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

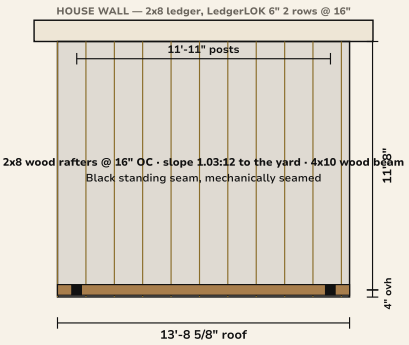
A-401 ROOF COVERS — PLAN + SECTION

Lean-to covers: ledger, rafters, beam, posts on the piers, clear heights, pitch, standing seam, Hardie, blinds

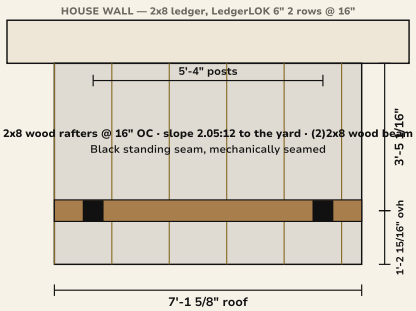
281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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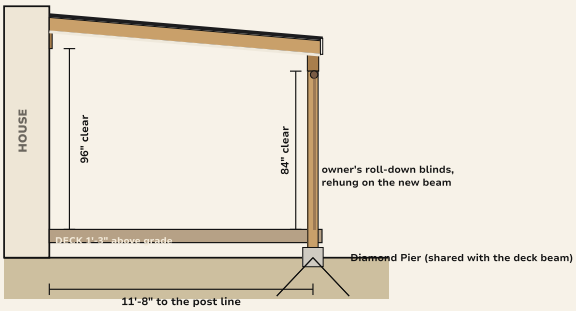
MAIN COVER — PLAN



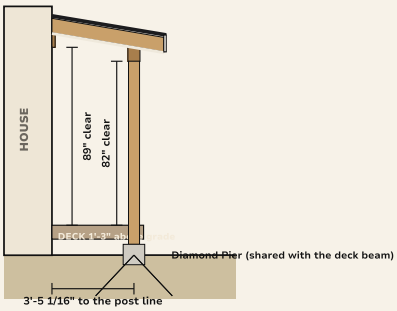
WALKWAY COVER — PLAN



MAIN COVER — SECTION



WALKWAY COVER — SECTION



Standing seam 24 ga, 16" panels, 1-1/2" mechanical (double-lock) seam — rated for 1.03:12 · high-temp ice & water underlayment over 7/16" OSB
Headwall flashing counterflashed into the siding · color-matched eave / rake trim · snow bar across the low eave
HardieSoffit cedarmill ceiling · HardieTrim fascia + rakes (Arctic White) · H2.5AZ at every rafter · post caps per D-503
6x6 Douglas fir rough posts on ABA66RZ bases on the Diamond Pier bolt · BC46Z cap to the 4x10 Douglas fir beam · the engineer's uplift hold-down governs

Standing seam 24 ga, 16" panels, 1-1/2" mechanical (double-lock) seam — rated for 2.05:12 · high-temp ice & water underlayment over 7/16" OSB
Headwall flashing counterflashed into the siding · color-matched eave / rake trim · snow bar across the low eave
HardieSoffit cedarmill ceiling · HardieTrim fascia + rakes (Arctic White) · H2.5AZ at every rafter · post caps per D-503
6x6 Douglas fir rough posts on ABA66RZ bases on the Diamond Pier bolt · AC6Z cap to the (2)2x8 beam · the engineer's uplift hold-down governs

Plans and sections not to a common scale · dimensions govern · the stamped set governs

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

A-402 Roof covers — exploded isometric, labeled

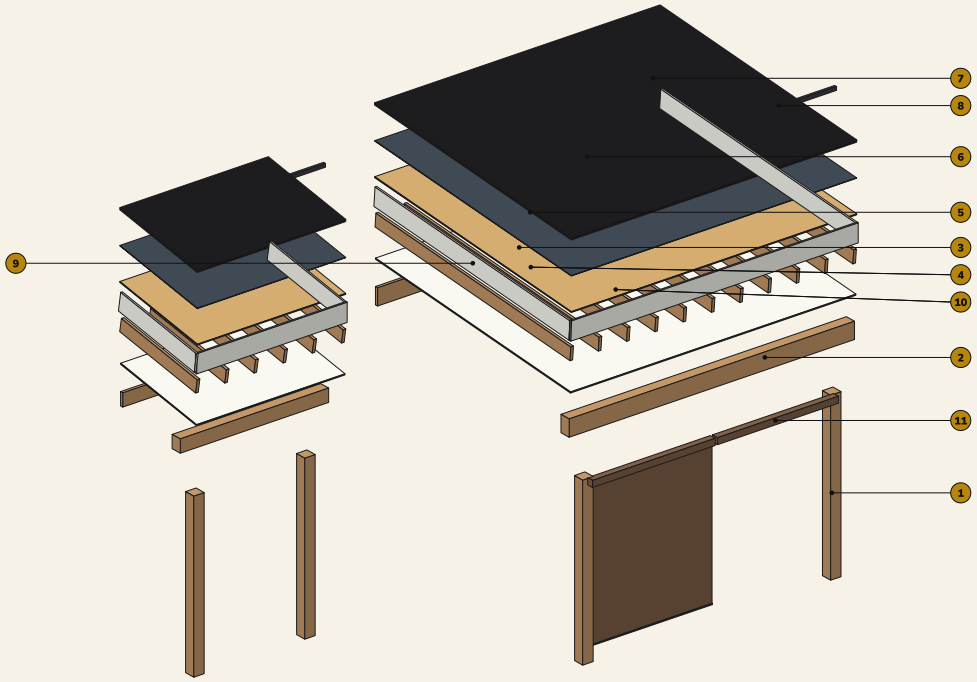
A-402

ROOF COVERS — EXPLODED ISOMETRIC, LABELED

Both lean-to covers pulled apart layer by layer

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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KEY

1. 6x6 Douglas fir rough cover post on ABA66RZ (shared Diamond Pier)
2. 4x10 cover beam, Douglas fir, BC46Z cap
3. 2x8 cover ledger, wood, LedgerLOK 6" 2 rows, headwall flashed
4. 2x8 rafters @ 16" OC, wood, H2.5AZ at the beam
5. 7/16" OSB roof sheathing
6. High-temp self-adhered underlayment
7. Black standing seam, 24 ga, mechanically seamed
8. Snow retention bar, clamped to the seams
9. HardieTrim fascia + rakes Arctic White
10. HardieSoffit cedarmill ceiling Arctic White
11. Existing roll-down blinds (stay)

Lifts are for clarity only. Order of work: posts on ABA66RZ bases, beams on their post caps, ledger on LedgerLOK, rafters with H2.5AZ, soffit, OSB, underlayment, panels, trims, snow bar.

Axonometric · not to scale · the model the plans are drawn from

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

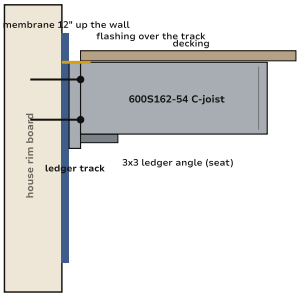
D-501 DETAILS — STEEL FRAME

Steel ledger · joist clips · boxed beam on the shared pier · Diamond Pier · deck edge · under the deck

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

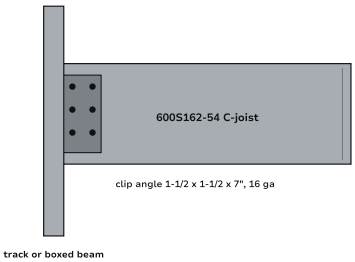
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1 STEEL LEDGER AT THE HOUSE



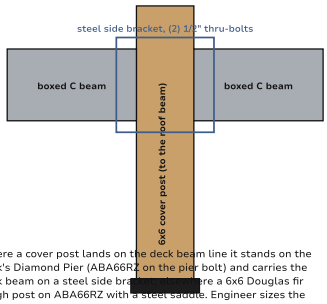
600T ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer confirms) · flashing and membrane as for wood

2 JOIST TO TRACK / BEAM



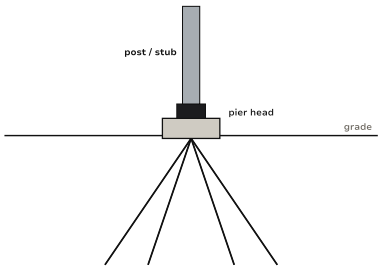
#12-14 self-drilling hex-washer screws, 4 per leg minimum (clip to joist web, clip to track / beam web) — the engineer's clip schedule governs. Cut ends and field holes cold-galvanized.

3 BOXED BEAM · SHARED PIER



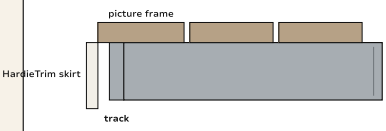
Where a cover post lands on the deck beam line it stands on the deck's Diamond Pier (ABA66RZ on the pier bolt) and carries the deck beam on a steel side bracket; above the deck it is a 6x6 Douglas fir rough post on ABA66RZ with a steel saddle. Engineer sizes the pier for roof + deck and the uplift hold-down.

4 DIAMOND PIER



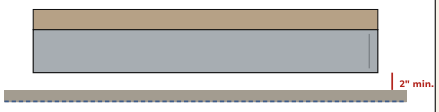
DP-50/50 (4 pins) — no excavation; frost-protected by the pins (submit the evaluation report). 811 locate first.

5 DECK EDGE · HARDIE SKIRT



TimberTech Vintage Weathered Teak: hidden fasteners in the field, the picture frame face-fastened with color-matched plugs — every screw the maker's steel-frame screw for 14–18 ga. HardieTrim skirt on furring screwed to the track.

6 UNDER THE DECK · LOW END



2" gravel over 6-mil poly, sloped away from the house

Keep 2" min. between the frame and the gravel.

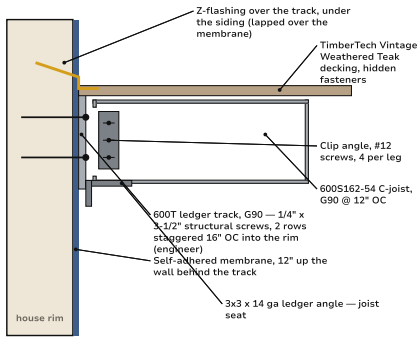
D-502 CONNECTION DETAILS — CEMCO STEEL DECK FRAME

Ledger track · joist clips · boxed beam · post on the pier · shared pier · deck edge · stair

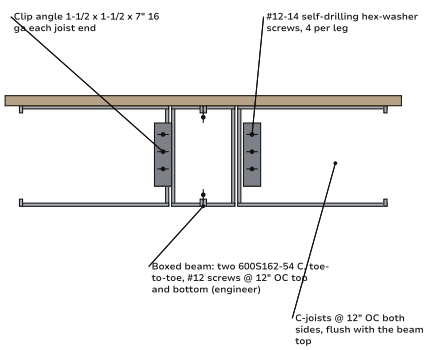
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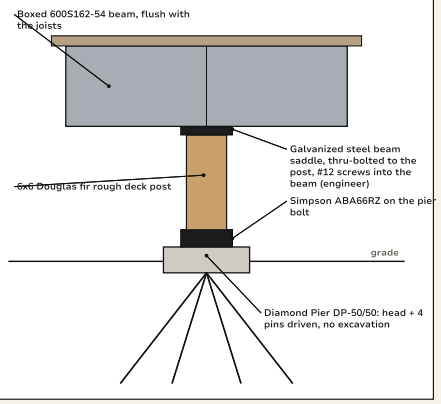
1 LEDGER TRACK AT THE HOUSE — SECTION



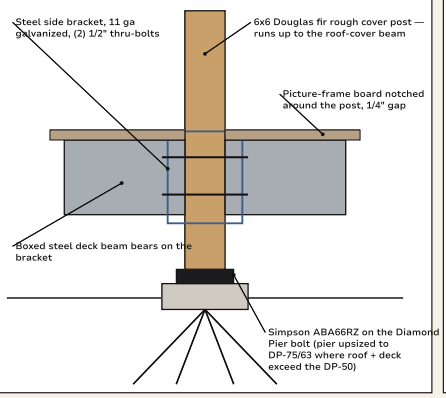
2 JOIST TO BOXED BEAM — SECTION



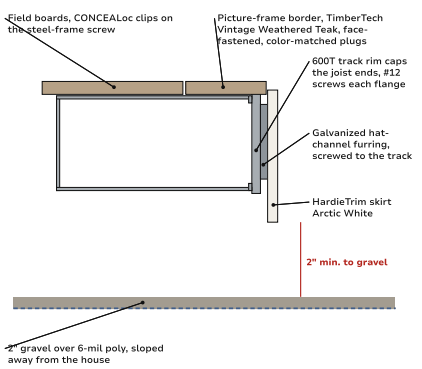
3 BEAM ON 6x6 POST + DIAMOND PIER



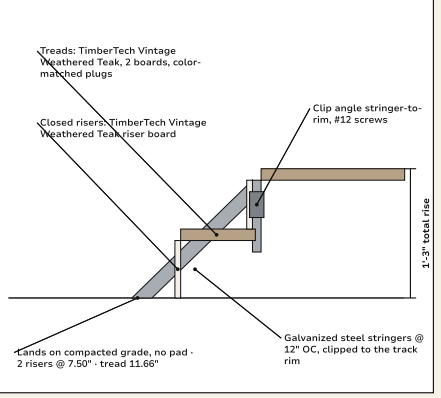
4 SHARED PIER — COVER POST + DECK BEAM



5 DECK EDGE — TRACK RIM, BORDER, HARDIE SKIRT



6 STAIR OFF THE MAIN DECK — SECTION



Schematic details, not to scale · the engineer of record's details and screw schedule govern

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

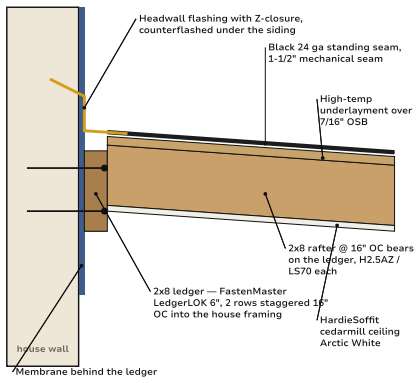
D-503 CONNECTION DETAILS — ROOF COVERS (WOOD FRAMING)

Ledger + headwall · eave at the beam · post base · rake · standing seam + clip · blinds

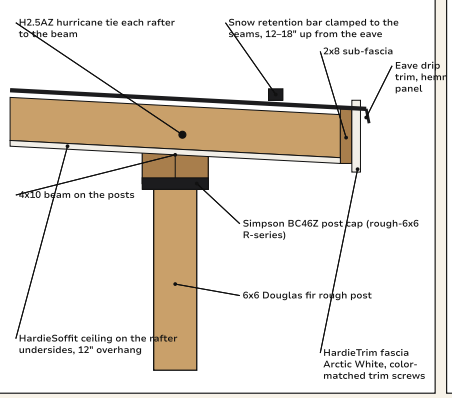
281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

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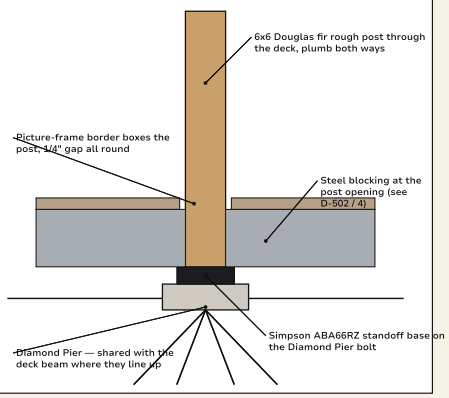
1 COVER LEDGER + HEADWALL — SECTION



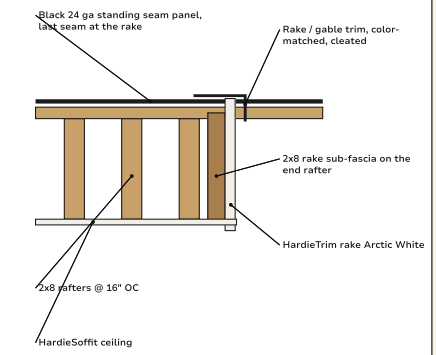
2 EAVE AT THE BEAM — SECTION



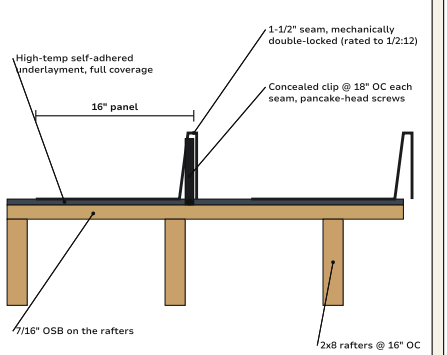
3 COVER POST BASE + DECK PENETRATION



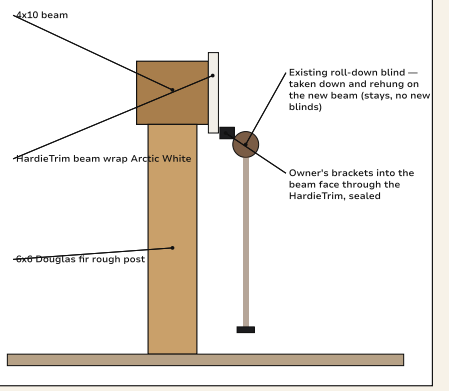
4 RAKE EDGE — SECTION ACROSS THE SLOPE

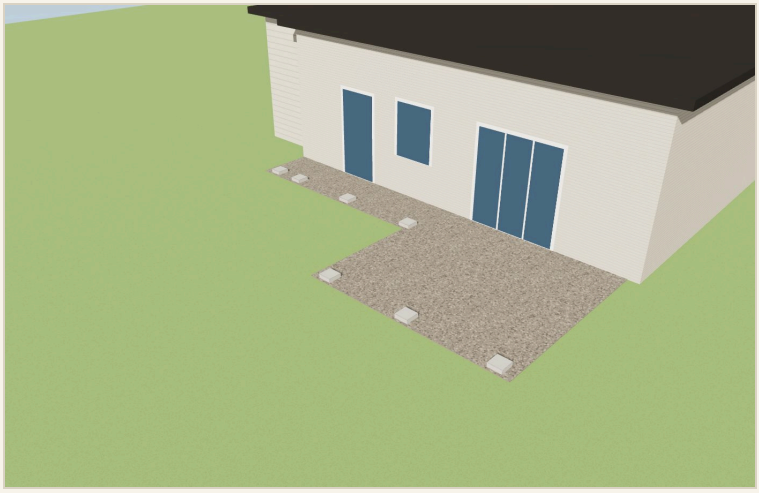


5 STANDING SEAM PANEL + CLIP



6 EXISTING BLINDS AT THE BEAM





Step 1 · Footings — 7 Diamond Pier DP-50/50 driven-pin footings

T-400 · Step 1 · Footings

GOES IN

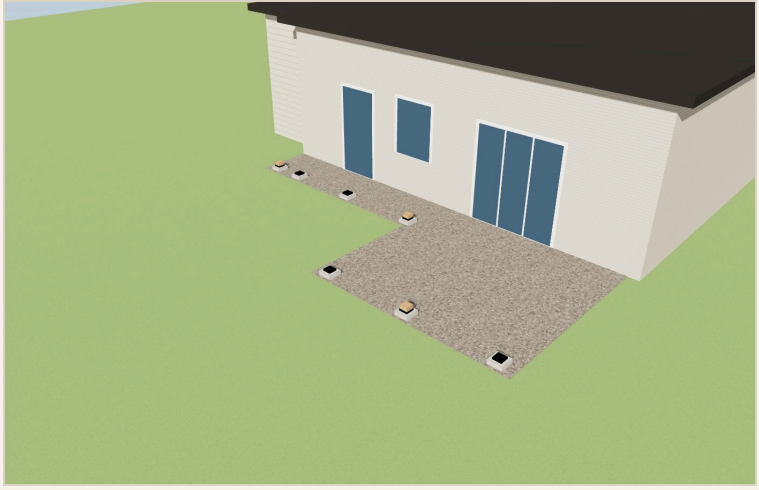
- (7) Diamond Pier DP-50/50 (head + 4 x 50" pins, plugs, caps)

HOW

- Call 811 and locate before any hole or pin. Strip the existing deck first.
- Pull the beam lines from the house wall: FLUSH boxed 600S162-54 W at 3'-6 3/8" from the reference wall, posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 13/16"; FLUSH boxed 600S162-54 M at 11'-9 3/8" from the reference wall, posts at 15'-3" / 21'-2 1/2" / 27'-2".
- Set 7 Diamond Pier heads level at grade on undisturbed soil and drive the 4 pins in each; caps on.

CHECK

- Footings on the layout within 1/2"; bearing on undisturbed soil
- Pins fully driven, heads level
- Photos before backfill



Step 2 · Posts — 6x6 Douglas fir rough deck posts on ABA66RZ bases, steel saddles to the beams; 6x6 Douglas fir rough cover posts carry the deck beams where they line up

T-401 · Step 2 · Posts

GOES IN

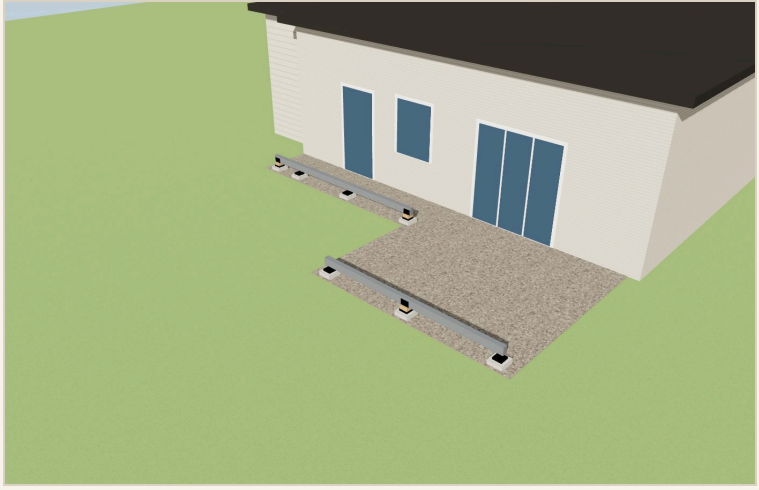
- (4) Steel side bracket, 11 ga galvanized — deck beam bolted to the 6x6 cover post (shared pier)
- (1) 6x6x8 Douglas fir rough — deck posts on the deck-only piers
- (3) Simpson ABA66RZ post base on the Diamond Pier bolt — deck post
- (3) Galvanized steel beam saddle, 11 ga — boxed steel beam to the 6x6 post top
- (2) Simpson ABA66RZ post base on the shared Diamond Pier bolt — cover post (pier counted with the deck) — Main cover
- (2) Simpson ABA66RZ post base on the shared Diamond Pier bolt — cover post (pier counted with the deck) — Walkway cover

HOW

- Deck-only piers: 6x6 Douglas fir rough posts on ABA66RZ bases, galvanized steel saddles on top carrying the boxed beams, cut so the beam top lands at the joist top (flush beams): FLUSH boxed 600S162-54 W 6" / FLUSH boxed 600S162-54 M 6" — field-measure each.
- Where a roof-cover post lands on the deck beam line it stands on the same Diamond Pier on an ABA66RZ base; the 6x6 Douglas fir rough post runs up to the cover beam and the deck beam bolts to it with the steel side bracket.
- Plumb both ways and brace until the beams are on.

CHECK

- Posts plumb to a string; tops on the beam line
- Every bracket and saddle fastened: thru-bolts and #12 screws per the engineer



Step 4 · Beams — FLUSH boxed 600S162-54 W (4 posts) · FLUSH boxed 600S162-54 M (3 posts)

T-403 · Step 4 · Beams

GOES IN

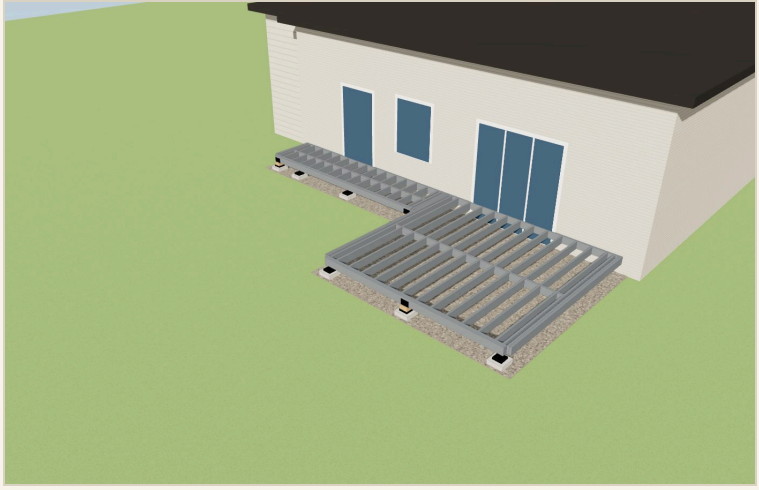
- (53) CEMCO 600S162-54 C, G90 — boxed beams (2 C toe-to-toe)
- (4) Steel side bracket, 11 ga galvanized — deck beam bolted to the 6x6 cover post (shared pier)

HOW

- FLUSH boxed 600S162-54 W: 14'-2 1/2" one piece, 4 posts @ 5'-9 13/16" max
- FLUSH boxed 600S162-54 M: 12'-4 1/2" one piece, 3 posts @ 5'-11 1/2" max
- Box two C sections toe-to-toe, #12 self-drilling screws per the engineer's pattern; set flush with the joist tops; brackets to the stubs and cover posts.

CHECK

- Beam top = joist top (flush)
- Brackets fully fastened; cut ends cold-galvanized



Step 5 · Frame — CEMCO 600S162-54 C-joists at 12" OC, 600T track rims, ledger track on a steel angle, clip angles, all G90

T-404 · Step 5 · Frame

GOES IN

- (377) CEMCO 600S162-54 C-joist (6", 54 mil / 16 ga, G90) — joists, PF joists, blocking
- (79) CEMCO 600T track (54 mil / 16 ga, G90) — rims (single ply, caps the joist ends) + ledger track
- (28) Steel ledger angle 3" x 3" 14 ga G90 — joist seats on the ledger track
- (1) Grace Vycor Deck Protector 12" x 75' (wall membrane behind ledger + side rims)
- (3) Aluminum L-flashing 10' (over ledger + side rims)
- (1) 6-mil poly sheeting 10' x 100' — under the decks
- (1.4) 3/4" washed gravel, 2" over the poly — under the decks (delivered)
- (60) Steel clip angle 1-1/2" x 1-1/2" x 7" 16 ga G90 — joist to track / beam
- (4) #12-14 x 1" hex-washer self-drilling screw, Climaseal, 500 ct
- (1) Structural screw 1/4" x 3-1/2" through the ledger track into the house rim board (or per the engineer)
- (8) Lateral-load tension tie for steel joists (Simpson DTT2Z w/ steel-joist kit or the engineer's detail) — IRC R507.9.2 equivalent

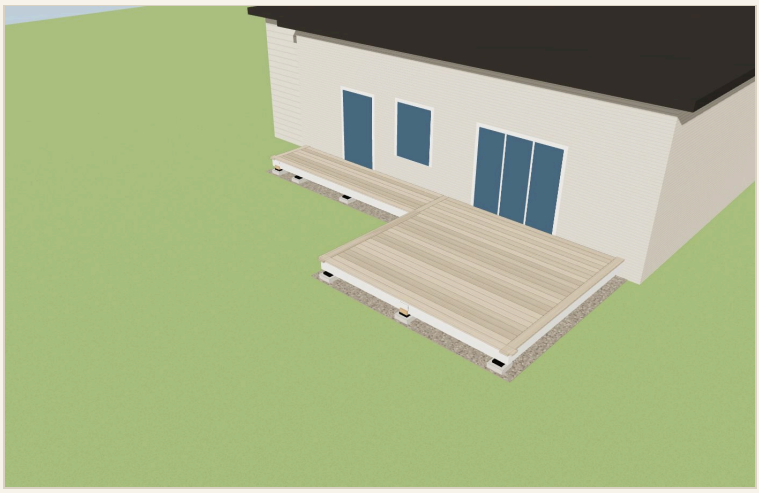
HOW

- 600T ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer confirms) · flashing and membrane as for wood
- W: 16 CEMCO 600S162-54 C-joists @ 12" OC x 3'-3 1/8"; M: 14 CEMCO 600S162-54 C-joists @ 12" OC x 11'-6 1/8". Joists seat on the ledger angle and clip to the ledger track and the boxed beams with clip angles, 4 #12 screws per leg.
- Track caps every joist end at the rims; picture-frame and breaker joists doubled where the borders land.
- Gravel over 6-mil poly under the frame before the decking (2" min. clear at the walkway low end).
- 600S162-54 C-joists @ 12" OC · single-ply 600T track rims · boxed 600S162-54 beams · all G90 · clip angles with #12 self-drilling screws (4 per leg) · cut ends cold-galvanized · no joist tape needed

CHECK

- Frame square: diagonals within 1/8"
- Every clip fully screwed; ledger screws per the schedule

- Cut ends cold-galvanized; photos of the ledger, clips and footings before decking



Step 6 · Decking — TimberTech Vintage Weathered Teak, Weathered Teak border, hidden fasteners

T-405 · Step 6 · Decking

GOES IN

- (31) TimberTech Vintage Weathered Teak 1x6x16 Grooved
- (4) TimberTech Vintage Weathered Teak 1x6x16 Square Edge — borders & dividers
- (1) TimberTech Vintage Weathered Teak 1x6x20 Square Edge — borders & dividers
- (5) Deck skirt — HardieTrim 4/4 x 12 x 12' smooth ColorPlus Arctic White (wraps the double rim, in place of the composite fascia)
- (1) Color-matched trim screws for HardieTrim, 100 ct (Arctic White)
- (1) ColorPlus touch-up kit, Arctic White
- (1) TimberTech CONCEALoc hidden fastener 500 ct (screws incl.)

HOW

- CONCEALoc at every joist in every gap (448)
- Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then alternate top/bottom every 12", pre-drilled · stair treads 2 per stringer per board
- Rows per panel: panel 1 (0"–14'-9") 7 rows · panel 2 (14'-9"–27'-8") 24 rows · 1 ripped pieces · 1 breaker. Borders and breakers first, then the field rows from the outer edge toward the house.

CHECK

- 3/16" gaps held
- No cut end shows at the outer edge
- Color: every board from the same run



Step 7 · (no rail)

T-406 · Step 7 · Rail

GOES IN

HOW

CHECK

- —
- Cables tensioned / balusters under 4" clear
- End of construction



Step 8 · Roof covers — 6x6 Douglas fir posts, 4x10 beam, 2x8 rafters, OSB, underlayment, standing seam, Hardie soffit / fascia / rakes

T-407 · Step 8 · Porch cover

GOES IN

- (2) 6x6x10 Douglas fir rough post (cover) — Main cover
- (1) 4x10x14 Douglas fir — cover beam — Main cover
- (1) 2x8x16 #1 SYP — cover ledger on the house — Main cover
- (10) 2x8x14 #1 SYP — cover rafters @ 16" OC — Main cover
- (3) 2x8x16 #1 SYP — cover fascia / sub-fascia — Main cover
- (6) 7/16" OSB roof sheathing 4x8 — Main cover
- (171) Standing seam roof — 24 ga steel 16" 1-1/2" mechanically seamed (double-lock, low-slope rated) panels x 12.5' (Black, PVDF), 10 panels — Main cover
- (1) High-temp self-adhered ice & water underlayment, 2 sq roll (under every metal panel) — Main cover
- (2) Standing seam eave / drip trim 10', color-matched — Main cover
- (3) Standing seam gable / rake trim 10', color-matched — Main cover
- (2) Headwall (ledger) flashing with Z-closure 10', color-matched — into the house siding / counterflash — Main cover
- (3) Foam + metal closure strips at eave and headwall, 10' — Main cover
- (1) Panel clips + pancake-head screws, box of 250 — Main cover
- (1) Color-matched #10 woodgrip trim screws w/ washers, 250 — Main cover

HOW

- Cover ledger on the house at the top of the pitch, LedgerLOK 2 rows; 6x6 Douglas fir rough posts on ABA66RZ bases on the Diamond Piers (shared with the deck beam where they line up), main cover: 4x10 beam on BC46Z caps / walkway cover: (2)2x8 beam on AC6Z caps; 2x8 rafters @ 16" OC with H2.5AZ ties and the overhang; 2x8 sub-fascia and OSB. Cover framing is wood (SYP 2x8 rafters, Douglas fir posts; Douglas fir 4x beam).
- High-temp self-adhered underlayment over every sheet, headwall flashing counterflashed into the house wall, eave trim, then the 24 ga snap-lock panels one piece eave to headwall — clips 18" OC on each seam, no face screws in the field; rake trim and closures last; snow retention bar clamped to

the seams 12-18" above the low eave, the full width.

- Ceiling: HardieSoffit 4x10 cedarmill Arctic White panels 6" OC stainless ring-shank at every rafter; HardieTrim 4/4 x 7.25 smooth Arctic White fascia and rakes over the 2x8 sub-fascia, color-matched trim screws.

CHECK

- Ledger flashed and counterflashed — hose test before the ceiling closes
- Posts plumb over blocking, beam level, rafters tied
- Panels straight, seams engaged full length, no oil-canning; snow bar continuous
- Water sheds clear of the deck edge

3D MODEL



The interactive 3D model lives at the online version of this document.

Prices: rate card · GC at cost · sold at $36,300 · 31.7% margin on the sold price (32.4% on the work) · financed ÷ 0.93 unless the job says otherwise. Internal — not for the homeowner.